Wind, slope, and the want of a tail: a computational feasibility study of Eilmer of Malmesbury's flight

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Abstract

William of Malmesbury records that in the early eleventh century a monk of his abbey, Eilmer, fixed wings to his hands and feet, launched from a tower, travelled "more than a furlong" (about 200 m), fell, broke both legs, and later blamed the fall on having no tail. This paper tests whether that account is consistent with elementary flight physics, using public data and methods at the level of an undergraduate mechanics course: a digital elevation model of the site, twenty years of reanalysis wind data, a slope-lift estimate, a point-mass glide integrated from a standing launch, a linear pitch-stability analysis, a Monte Carlo sample over the unknown apparatus and launch day, and an impact model for the legs. A glider leaving the tower at walking pace must lose about 11.5 m of height to reach its 15.1 m s^{-1} flying speed, so a long flight needs a strong wind. Flown out over the valley into that wind, none of 40 000 sampled cases covers 200 m. Flown along the southwest-facing slope in ridge lift, 6.9 % do (95 % confidence interval 6.6–7.1 %), with a median launch wind of 7.6 m s^{-1}. Without a tail the wing diverges in pitch, with a median doubling time of 0.29 s, comparable to human reaction time, during a flight of about 24 s; a tail removes the divergence. A controlled arrival, even without a flare, does not break the legs; an arrival after loss of pitch control often does. The account is physically possible but marginal, and the missing tail is a sufficient cause of the fall. Each step is posed as a student exercise, with worked answers.

1 Introduction

William of Malmesbury, writing his *Gesta Regum Anglorum* in about 1125, records that Eilmer, an older monk of his own abbey, had in his youth fixed wings to his hands and feet, launched from a tower, and travelled "more than a furlong" before he fell, broke both legs, and was lame for the rest of his life [1]. Eilmer, William adds, used to say that he fell because he had forgotten to fit a tail. Lynn White's study of the passage established it as the earliest credible account of a gliding attempt in Western Europe [2].

The account is short, but it states a launch point, a distance, a wind, a failure, an injury, and a cause. Each of these can be checked against physics that is taught to undergraduates. This paper asks whether the event William describes is physically possible at Malmesbury, and whether a missing tail is enough to explain the fall. It does not ask whether Eilmer understood aerodynamics, and it does not try to reconstruct his apparatus, which no source describes beyond its wings. The unknown apparatus is instead treated as a range of plausible parameters and sampled.

The problem makes a compact teaching case for undergraduate mechanics. It assumes calculus, Newtonian mechanics, and elementary probability, and suits students from the second

year onwards and their instructors. Answering it needs the energy balance of a glider at launch, flight in a moving air mass, lift over a slope, the neutral-point criterion for pitch stability, the growth rate of an unstable mode, Monte Carlo estimation with confidence intervals, and the mechanics of a feet-first impact. Each step uses public data and a few lines of code, and each produces a number that can be compared with the historical text. Section 6 sets the steps as exercises.

Section 2 summarises the source and the site, Section 3 the models, Section 4 the results, and Section 5 what they do and do not show.

2 The source and the site

William was a monk of Malmesbury writing about a century after the flight, and he could have heard the story from men who knew Eilmer [2, 3]. His Latin and a standard translation are given in Supplementary S1. He says that Eilmer "collected the breeze on the summit of a tower", that the violence of the wind and the swirling of the air brought him down, and that he broke both legs. The date is uncertain: the traditional estimate is about 1000–1010, and Aitcheson [4] argues it could be as late as the early 1040s. The analysis below samples launch weather from a modern climatology, so it does not depend on the year.

Earlier and later accounts of human flight attempts are mostly legend [5]. The one other early account with a stated device is that of Abbas ibn Firnas in ninth-century Córdoba, preserved in later Andalusian chronicles, which describes a structure of feathers and silk and a bad landing also blamed on the lack of a tail [2, 6]; Hallion [7] reads it as closer to a parachute descent. Neither source gives dimensions, materials beyond these, or how the wings were attached. There is therefore no historical basis for a specific design, and the models here sample wing area, mass, wing shape, and drag over wide ranges instead (Supplementary Table S1).

The abbey stands on a narrow spur between two branches of the Bristol Avon (Fig. 1). The present church is not Eilmer's, and his tower's position and height are unknown. The analysis therefore launches from the abbey site and samples the tower height between 15 m and 25 m, around White's estimate of about 18 m [2]. Woosnam [8] argues that the flight went down the steep slope southwest of the abbey towards the river, which is the side shown in Fig. 1.

3 Methods

3.1 Terrain

The terrain is a digital elevation model (DEM) of the abbey spur assembled from the Amazon Web Services (AWS) Terrain Tiles at zoom level 14, a grid spacing of about 6 m. Slope and aspect (the compass direction a slope faces) are computed within 400 m of the abbey, and an elevation section is taken along the southwest–northeast axis. A 6 m grid smooths the steep edge of the spur, so the computed slopes are lower bounds; airborne laser scanning (LIDAR) would resolve it.

3.2 Wind climatology

Hourly wind at 10 m height at the abbey for 2004–2023 (175 320 hours) is taken from the ERA5 reanalysis of the European Centre for Medium-Range Weather Forecasts [9], through the Open-Meteo archive [10]. An hour counts as favourable for launch when the wind blows from the southwest sector (191°–259°, centred on the slope aspect) at 5–12 m s^{-1}. Fractions are reported for the year and for each season, with 95 % confidence intervals (CI) from the spread



Figure 1: Malmesbury Abbey seen from the southwest. The view looks northeast from the water meadows of the Avon, about 300 m west-southwest of the abbey, up the southwest-facing slope that the prevailing wind strikes. The present church is a twelfth-century building; the Anglo-Saxon church and tower from which Eilmer launched do not survive, so the launch point is taken as the abbey site and the tower height is sampled at 15–25 m. Photograph: Dr Duncan Pepper, 2012, via Geograph and Wikimedia Commons, licensed CC BY-SA 2.0.

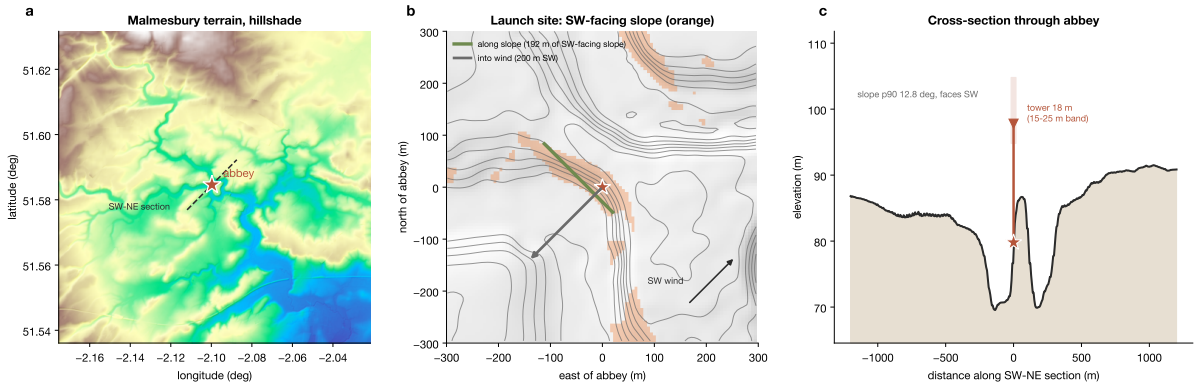


Figure 2: Terrain at Malmesbury Abbey. (a) Hillshaded DEM (AWS Terrain Tiles, about 6 m grid); the dashed line is the section in (c). **(b)** The launch site within 300 m of the abbey (star), with ground that faces southwest and is steeper than 5° in orange. The green line is the track along the slope, the longest straight run of that ground (192 m); the grey arrow is the first 200 m of the track into the wind. **(c)** Elevation along the southwest–northeast section, with the sampled tower heights. The 6 m grid smooths the edge of the spur, so the slopes are lower bounds.

across the twenty years. Modern data stand in for the eleventh century. Reconstructions of the North Atlantic Oscillation (NAO), the pressure pattern that sets the strength of the westerlies over Britain, disagree on its phase during the Medieval Climate Anomaly, but none indicates a long negative phase with weak westerlies [11, 12].

3.3 Ridge lift

Wind blowing up a slope is deflected upwards. To first order the vertical air velocity follows the surface gradient and decays with height,

$$w(x, z) = U \frac{\mathrm{d}h}{\mathrm{d}x} e^{-kz}, \tag{1}$$

where U is the wind speed, $h(x)$ the terrain height along the wind, z the height above the surface, and $k = 2\pi/L$ with L the horizontal scale of the slope. Equation (1) is applied to the measured section (Supplementary Fig. S1). The glide model uses the simpler surface estimate $w = \eta U \sin\theta_s$, with θ_s the slope angle and $\eta \le 1$ an efficiency that allows for flow that does not follow the surface. The lift acts only over the slope: for 40 m on the track that heads into the wind, where the section flattens onto the valley floor, and for 192 m on the track that follows the slope, the longest continuous stretch of southwest-facing ground steeper than 5° (Fig. 2b).

3.4 Glide performance

The glide ratio L/D of an unknown wing is estimated from the parabolic drag polar $C_D = C_{D0} + C_L^2/(\pi e A)$, whose maximum is

$$(L/D)_{\max} = \frac{1}{2}\sqrt{\frac{\pi e A}{C_{D0}}}, \tag{2}$$

where A is the aspect ratio (span squared over area), e the span efficiency, and C_{D0} the zero-lift drag coefficient [13]. Each factor can be bounded for a cloth wing carrying a prone man. The aspect ratio is taken as 3–5, a span of about one to two arm spans for areas of 5–10 m^2. Span efficiencies of 0.6–0.8 cover untwisted, non-elliptical wings [13]. The zero-lift drag $C_{D0} = 0.05$–0.15 combines the profile drag of a cambered membrane at a Reynolds number near 10^6 [14, 15] with the drag of the pilot's body, frame, and bracing [16]. These ranges give $(L/D)_{\max} = 3.1$–7.9, with 4.7 at their centres. The glide calculation flies every wing at its best glide ratio, which favours the account.

The trim airspeed of a wing of area S and mass m at lift coefficient C_L is $V = \sqrt{2mg/(\rho S C_L)}$, which is 15.1 m s^{-1} for the nominal configuration of Fig. 3 (80 kg, 7 m^2, $C_L = 0.8$). Figure 3 shows the centre of the sampled ranges and is not a reconstruction; the only historical detail it carries is the wing fixed to the hands and feet, with no tail.

3.5 Launch and glide trajectory

A man stepping off a tower does not start at flying speed. He leaves at about 2 m s^{-1} relative to the ground and must dive until the airspeed reaches V. The flight is therefore integrated in time as a point mass held at constant C_L,

$$\dot{V} = g\sin\gamma - \frac{D}{m}, \qquad \dot{\gamma} = \frac{1}{V}\left(g\cos\gamma - \frac{L}{m}\right), \qquad \dot{z} = -V\sin\gamma + w, \tag{3}$$

where V is now the airspeed, γ the descent angle below the horizontal, $L = \tfrac{1}{2}\rho V^2 S C_L$, and $D = L/(L/D)$. The airspeed at release is the jump speed plus the headwind. In still air the craft dives, gains speed, and pulls out in a damped oscillation, the phugoid.

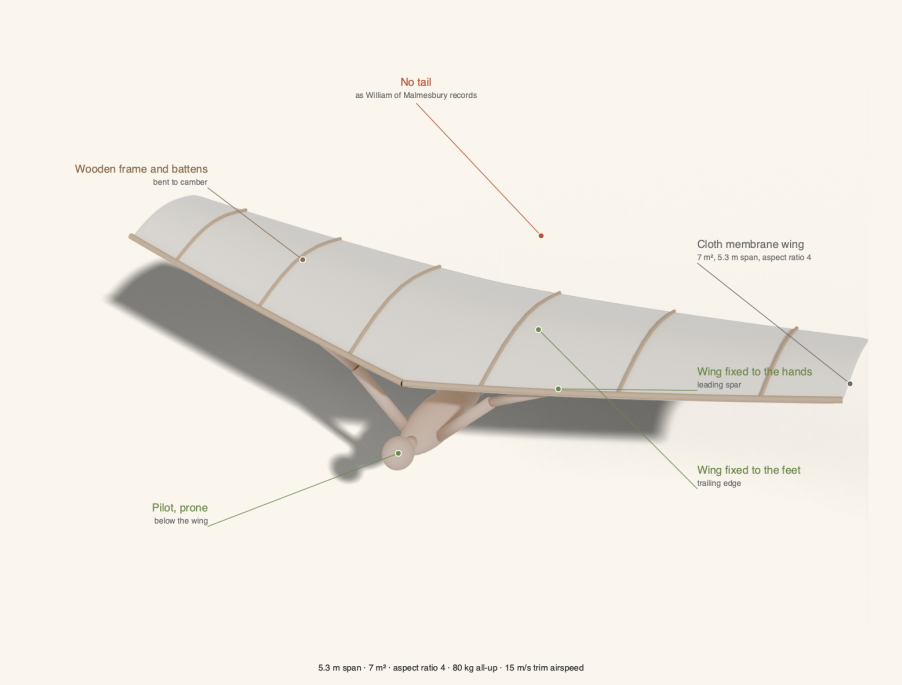


Figure 3: Nominal model configuration. A cloth wing on a wooden frame with the pilot prone beneath it, the wing fixed to his hands at the leading edge and to his feet at the trailing edge, and no tail. The dimensions are the centres of the sampled ranges (Supplementary Table S1), not a historical reconstruction: no source describes the apparatus beyond its wings.

The wind also changes the ground covered, and two tracks bracket the possibilities. On the first, the pilot heads southwest into the wind, down the slope towards the river [8]; his ground speed is $V\cos\gamma - U$, and the ground is the measured section. On the second, he flies northwest along the slope with the wind from the side and crabs, heading partly into the wind so that it does not drift him off the track; his ground speed is $\sqrt{(V\cos\gamma)^2 - U^2}$, and the ground is taken as a contour 5.1 m below the abbey, half-way down the slope. The second is how hang-glider and sailplane pilots use ridge lift [17]. William's "more than a furlong" is tested as a ground range of at least 200 m.

3.6 Pitch stability

A wing holds its pitch attitude on its own if the centre of gravity (CG) lies ahead of the neutral point, the point about which the pitching moment does not change with angle of attack. Both are written as fractions of the mean chord c from the leading edge, x_{cg} and x_{np}. A cambered wing without a tail has $x_{\mathrm{np}} \approx 0.25$; a tail moves it aft, to 0.53 for the nominal tail, and adds pitch damping [13].

The pilot's CG is set by his body. With the wing fixed to his hands at the leading edge and his feet at the trailing edge, his mass centre lies between them. Standing, a man's mass centre is at about 55 % of his height above the soles [18], and his reach from soles to hands raised overhead is about 1.25 times his height. Laid out from hands to feet, the mass centre therefore sits at about 0.55 of that distance behind the hands. The sampled range $x_{\mathrm{cg}} = 0.35$–0.55 runs from this straight posture forward to a pilot who draws his hips up under the leading edge. The pitch analysis is also run across $x_{\mathrm{cg}} = 0.20$–0.60, to show where the conclusion would change.

The short-period pitch motion is linearised in angle of attack and pitch rate with standard stability derivatives [13]. When the CG is aft of the neutral point, the state matrix has a positive

real eigenvalue λ, and a disturbance doubles in $t_2 = \ln 2/\lambda$. t_2 is compared with the duration of the flight and with the effective time delay of a human operator in a manual control task, about 0.15–0.3 s [19]. The nonlinear response to an 8° gust is also integrated, with control taken as lost beyond a pitch attitude of 60°.

3.7 Monte Carlo sample

The apparatus and the day are unknown, so 40 000 cases are drawn from uniform ranges (Supplementary Table S1) with a fixed random seed. The draws are wing area, mass, trim C_L, aspect ratio, span efficiency, zero-lift drag, tower height, slope angle, lift efficiency η, CG, and the neutral-point shift from a tail. The glide ratio follows from Eq. (2) and is not drawn directly. The launch wind is drawn from the measured ERA5 southwest-sector hours at 3–16 m s^{-1}, on the assumption that Eilmer chose a windy day. Each case is flown on both tracks with Eq. (3) and scored on whether it covers 200 m, and whether t_2 is shorter than the flight, with and without a tail. Fractions carry Wilson score 95 % confidence intervals.

3.8 Impact

William records that Eilmer broke both legs and survived. In a feet-first landing the body stops over a distance d of leg flexion and ground give, so the mean force for a vertical arrival speed v is

$$F = \frac{mv^2}{2d}, \qquad \text{shared between two legs.} \tag{4}$$

The tibia breaks under axial and combined loads of 7.5–11.3 kN per leg [20–24], which sets a fracture-onset speed $v^\star = \sqrt{2dF_{\mathrm{tot}}/m}$ for d = 10–35 cm. Two arrivals are compared for the cases that reach 200 m. In a controlled glide that ends without a flare, the vertical speed is the one at which Eq. (3) meets the ground. After a loss of pitch control the craft strikes at airspeed V along a path 35°–70° below horizontal, with vertical speed $V \sin\theta$. Equation (4) uses the mean force, so real peak loads are higher and the onset speeds are upper bounds.

4 Results

4.1 Site and wind

The abbey stands 10.2 m above the valley floor on a spur whose flanks face southwest (aspect 230°), with a mean slope of 5.2° and a 90th-percentile slope of 12.8° within 400 m (Fig. 2). The southwest-facing ground near the abbey forms a strip 192 m long running northwest from the abbey; heading southwest, the ground reaches the valley floor within 40 m.

Over twenty years the wind blows from the southwest sector at 5–12 m s^{-1} in 15.6 % of hours (95 % CI 14.3–17.0 %), about 1 370 hours a year, and in 21.1 % of winter hours (Fig. 4). Strong southwest winds are rarer: 6 m s^{-1} or more in 10.3 % of hours (906 hours a year), and 8 m s^{-1} or more in 3.4 % (CI 2.9–3.9 %; 296 hours a year).

The linear model (Eq. (1)) gives a peak upward air speed of 1.5 m s^{-1} just southwest of the abbey in an 8 m s^{-1} wind (Supplementary Fig. S1), close to the surface estimate $U \sin\theta_s$ = 1.8 m s^{-1} at the 90th-percentile slope. The still-air sink rate of the nominal craft at its best glide is $V/\sqrt{1 + (L/D)^2}$: 1.9 m s^{-1} at L/D 7.9 and 3.1 m s^{-1} at L/D 4.7. Ridge lift can therefore roughly cancel the sink of the better wings in a strong wind, and not that of the poorer ones.

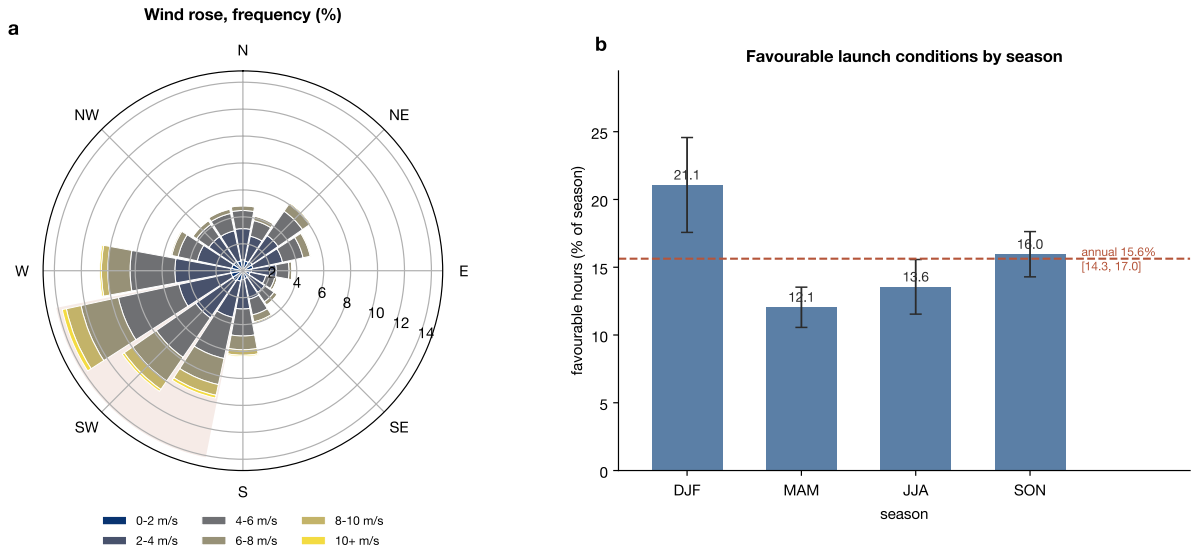


Figure 4: Wind at the abbey, 2004–2023. (a) Wind rose of ERA5 hourly 10 m wind (175 320 hours). **(b)** Fraction of hours from the southwest sector (191°–259°) at 5–12 m s^{-1}, for the year and by season (DJF, December–February; MAM, March–May; JJA, June–August; SON, September–November), with 95 % confidence intervals across the twenty years.

4.2 Launch and glide

In still air, a steady glide from the 18 m tower would cover 200 m only with L/D 11.1, beyond the range of Eq. (2). A standing launch does much worse. Reaching 15.1 m s^{-1} from 2 m s^{-1} costs $(V^2 - v_0^2)/2g \approx 11.5$ m of height, and the best wing, L/D 7.9, lands 14 m from the tower instead of the 142 m of a steady glide (Fig. 5b). A headwind supplies airspeed at release without the dive, which is why the wind matters for the launch as well as for the lift.

Into the wind, the headwind that eases the launch also slows the craft over the ground, and the lift ends within 40 m. None of the three wings in Fig. 5a reaches 200 m on this track at any wind up to 14 m s^{-1}; the longest flight is 110 m (dashed lines). Along the slope, the lift lasts for 192 m, and a crabbing craft can stay in it. The best wing reaches 200 m from a 18 m tower above 5.7 m s^{-1} of wind and the central wing above 10.6 m s^{-1}; the poorest wing does not reach it at any wind (Fig. 5a, solid). The threshold for the best wing moves from 3.8 m s^{-1} to 6.6 m s^{-1} as the tower height falls from 25 m to 15 m.

4.3 Pitch stability

Without a tail the neutral point is at 0.25 c, ahead of every sampled CG, so the wing is statically unstable in pitch (Fig. 6). Pitch damping delays the divergence until the CG is aft of 0.30 c. Across the sampled CG range the doubling time falls from 0.60 s at 0.35 c to 0.16 s at 0.55 c (0.12–0.89 s when the pitch inertia is halved or doubled); a 200 m flight of the nominal craft lasts about 16 s. After an 8° gust the tailless craft passes 60° of pitch in 0.74 s. With a tail the same gust dies away.

4.4 Monte Carlo sample

Into the wind, none of the 40 000 cases reaches 200 m (95 % upper bound 0.01 %); the median range is 62 m (Fig. 7a). Along the slope, 2 748 cases do, 6.9 % (CI 6.6–7.1 %), against a median range of 83 m and a 90th percentile of 180 m. The cases that reach it share a strong wind and

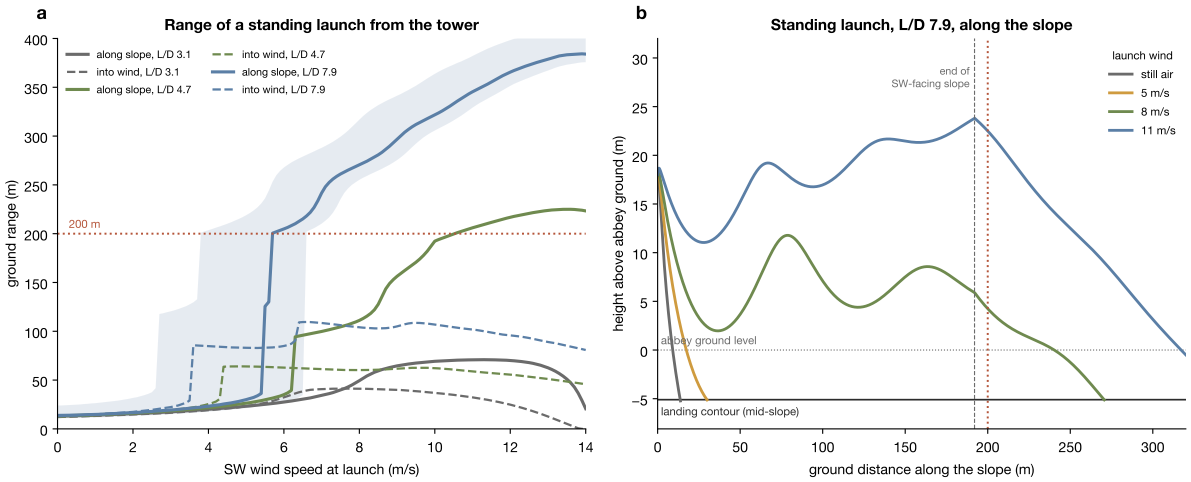


Figure 5: Range of a standing launch from the tower. **(a)** Ground range against launch wind for the nominal craft (Fig. 3) launched from 18 m at 2 m s^{-1}, for the lowest, central, and highest glide ratios of the drag-polar range, on the track along the slope (solid) and into the wind (dashed). The shaded band is the best wing along the slope for towers of 15–25 m. The step near 5.5 m s^{-1} is where ridge lift first matches the sink of the best wing. **(b)** Trajectories along the slope for the best wing (L/D 7.9). In still air the craft dives to gain speed and lands 14 m out; in wind it climbs and falls in the lift and descends once past the end of the southwest-facing slope.

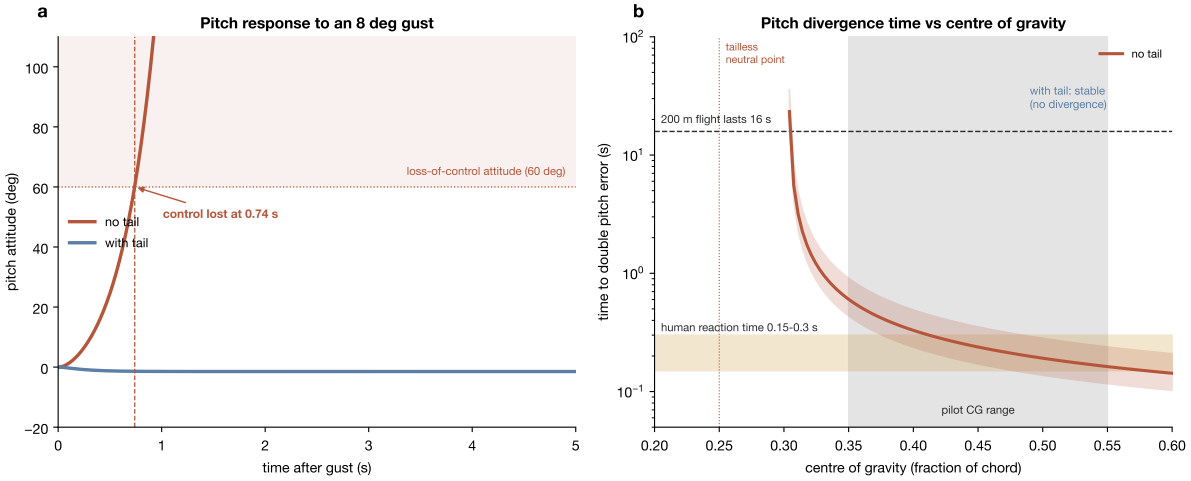


Figure 6: Pitch stability with and without a tail. **(a)** Pitch attitude after an 8° gust for the nominal craft with its CG at mid-range. Without a tail the attitude passes 60° after 0.74 s; with a tail the disturbance damps out. **(b)** Time for a pitch disturbance to double against CG position for the tailless craft; the band is pitch inertia from half to twice nominal. The grey band is the sampled CG range and the tan band the effective time delay of a human operator [19]. The tailed craft does not diverge at any CG shown.

a good wing: their median launch wind is 7.6 m s^{-1} (90 % above 5.4 m s^{-1}), and their median glide ratio is 5.8, against 4.7 over all cases. 34 % of all cases land within 50 m of the tower.

Among the cases that reach 200 m, the tailless doubling time has a median of 0.29 s (99th percentile 1.6 s) and the flight a median of 24 s (Fig. 7b). In every one of them the doubling time is shorter than the flight, and with a tail none diverges.

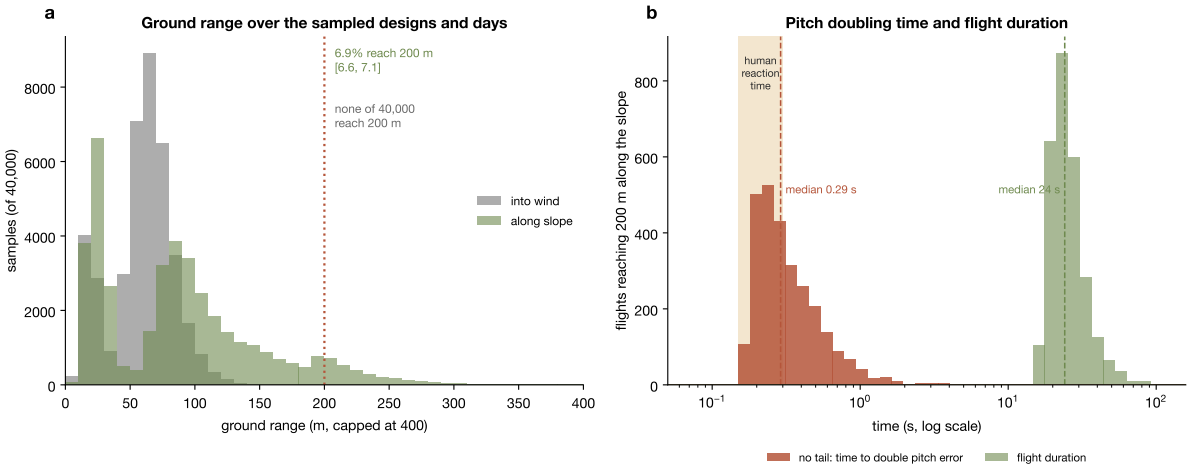


Figure 7: Monte Carlo sample of 40 000 apparatus and launch-day cases. Parameters are drawn from Supplementary Table S1, and the launch wind from the ERA5 southwest-sector hours. **(a)** Ground range on the two tracks. Along the slope 6.9 % of cases reach 200 m (95 % CI 6.6–7.1 %); into the wind none does. **(b)** For the 2 748 cases that reach 200 m along the slope, the time for a pitch disturbance to double without a tail, and the duration of the flight. The tan band is the effective time delay of a human operator.

4.5 Impact

For $d = 20$ cm the legs break above about 9.7 m s^{-1} of vertical speed, and the onset ranges from 6.1 m s^{-1} to 14.1 m s^{-1} across the stopping distances and fracture loads (Fig. 8a). A controlled glide that ends without a flare meets the ground at a median 2.2 m s^{-1} and at most 4.1 m s^{-1} of vertical speed, below the lowest onset speed, so it does not break the legs even without a flare (Fig. 8b). An arrival after loss of pitch control, at airspeeds of 10.3–19.4 m s^{-1}, has a median vertical speed of 10.1 m s^{-1}; 58 % of such arrivals exceed the central onset speed. The median deceleration over 20 cm is about 26 g, which feet-first falls are commonly survived.

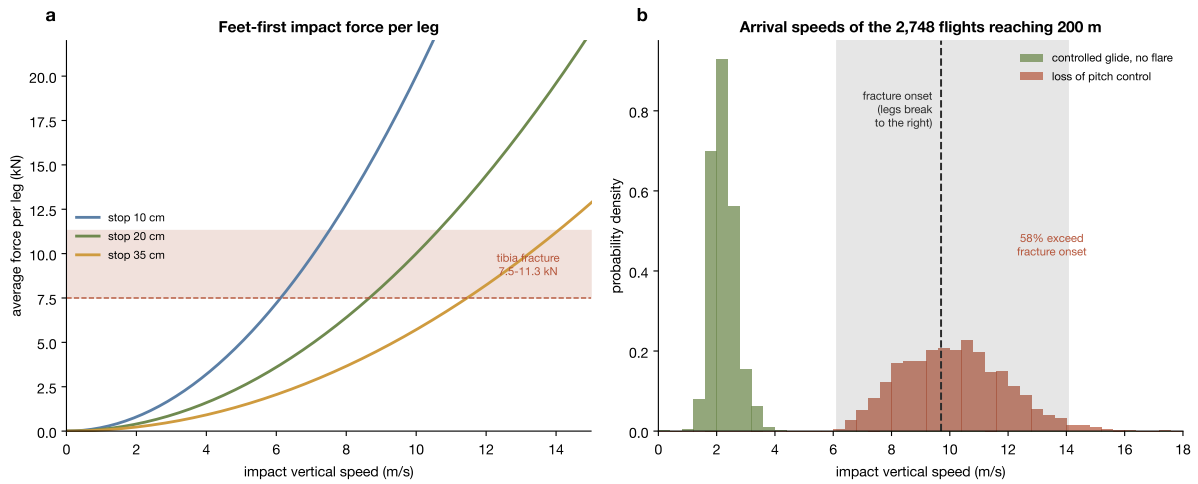


Figure 8: Arrival speed and leg fracture. (a) Mean feet-first impact force per leg against vertical speed for stopping distances of 10–35 cm, with the tibia fracture range (7.5–11.3 kN per leg). **(b)** Vertical arrival speeds of the 2 748 cases that reach 200 m, for a controlled glide that ends without a flare and for an arrival after loss of pitch control ($V \sin\theta$, $\theta = 35°$–$70°$). The grey band is the range of fracture-onset speeds and the dashed line its centre.

5 Discussion

5.1 What the account requires

Read against the models, William's account describes a narrow set of conditions. A flight of a furlong from a tower of this height needs a strong southwest wind, a wing near the better end of what cloth and timber allow, and a track that stays over the slope. Flying straight out over the valley, the route Woosnam [8] proposed, does not reach 200 m in any sampled case, because the headwind slows the craft over the ground and the lift ends at the foot of the slope. Flying along the slope reaches it in 6.9 % of cases. Southwest winds of 8 m s^{-1} or more blow for about 296 hours a year, so a man who waited for one would not have waited long. William's phrase that Eilmer "collected the breeze" fits a launch timed to such a wind.

Without a tail the craft diverges in pitch in a fraction of a second, but the doubling time is close to human reaction time rather than far below it. A pilot who moves his body could therefore hold the attitude for a while by continual correction, and a flight of tens of seconds followed by a loss of control is consistent with the model. The account names the trigger: the violence of the wind and the swirling of the air. A gust large enough to need a correction faster than his reactions would end the flight in under a second (Fig. 6a). The models do not show that he lost control, only that he would have been unlikely to keep it and that a tail would have removed the need to.

The broken legs are consistent with a loss of control but do not prove one. A controlled glide into the ground without a flare arrives too slowly to break the legs in this model, so the injury requires a fast, steep arrival. A loss of pitch control gives that, but so would a stall at height or a dive into the ground that is not caused by the missing tail.

5.2 Which results depend on the assumptions

Three results follow from measurement or from simple bounds and would survive large changes to the model. The slope faces the prevailing wind, and strong southwest winds are measured to occur (Figs. 2 and 4). The launch dive costs $(V^2 - v_0^2)/2g$ of height whatever the wing, so a

still-air flight of a furlong is not possible from this tower. A tailless wing is unstable in pitch whenever the CG is aft of about $0.30\,c$, and for a man fixed to the wing at his hands and feet the CG is near mid-chord. This last result holds if the sampled CG range brackets the real posture; Fig. 6b shows it would fail only if the pilot's mass centre lay ahead of $0.30\,c$, within the front of the chord near his hands.

The fraction of cases reaching 200 m depends on the ranges in Supplementary Table S1, and should be read as the share of the sampled ranges that works, not as a probability that the flight happened. It is most sensitive to the glide ratio and the strength and length of the ridge lift. The glide ratio rests on Eq. (2) with bounds on aspect ratio, span efficiency, and drag, and every wing is flown at its best glide, which favours the account. The lift rests on a first-order model over a smoothed 6 m terrain grid; the real slope is steeper than the grid shows, which would raise the lift, while turbulence and separation behind the spur edge would lower it. The point-mass model holds C_L fixed, so it does not let the pilot trade height for speed more efficiently than a stable glider would, and the pitch analysis treats the longitudinal motion alone. A large-eddy simulation over LIDAR terrain and a six-degree-of-freedom model with a pilot model would narrow these margins.

5.3 Limits of the comparison

The comparison with the text assumes that a furlong means about 200 m of ground, that the tower stood at or near the abbey, and that William reports what Eilmer said. A shorter flight would be easier to explain; a tower on lower ground would be harder. None of the models tests whether the structure could carry the flight loads, which would be a separate failure mode.

6 Using the study in teaching

The analysis divides into exercises that each use one topic from a first or second year mechanics course and one public data set or short script. They can be set singly, as problems, or together as a project in which students rebuild a figure and then change one assumption. Worked answers are in Supplementary S4.

1. **Energy at launch.** A glider leaves a tower of height H at $v_0 = 2\,\mathrm{m\,s^{-1}}$ and needs $V = 15.1\,\mathrm{m\,s^{-1}}$ to fly. Ignoring drag, find the height lost in reaching V. How large a headwind makes the launch dive unnecessary? Integrate Eq. (3) numerically in still air and compare the range with the steady glide estimate $H\,(L/D)$.

2. **Glide ratio from a drag polar.** Derive Eq. (2) by minimising C_D/C_L. Evaluate it at the corners of the ranges in Supplementary Table S1. Which parameter does the glide ratio depend on most, and which measurement on a model wing would narrow it?

3. **Flight in a moving air mass.** Draw the velocity triangle for a craft of airspeed V heading into a wind U, and for one crabbing along a track perpendicular to it. Show that the ground speeds are $V - U$ and $\sqrt{V^2 - U^2}$. Explain why a headwind helps the launch and shortens the range on the first track.

4. **Ridge lift and sink.** The upward air speed over a slope of angle θ_s is about $U \sin\theta_s$. The still-air sink rate is $V/\sqrt{1 + (L/D)^2}$. Find the wind at which the two are equal for L/D 4.7 and L/D 7.9 on a slope of 12.8°, and compare the answers with the ERA5 wind distribution.

5. **Static and dynamic pitch stability.** Write the two-state short-period model in angle of attack and pitch rate. Show that a positive real eigenvalue appears when the CG moves aft of a point that depends on the pitch damping, and compute the doubling time for the nominal craft at $x_{\mathrm{cg}} = 0.45$. Compare it with your own reaction time, measured with a ruler-drop test.

6. **Monte Carlo estimation.** Rerun the Monte Carlo with a different seed and sample size. Compute the Wilson interval for 2 748 successes in 40 000 trials by hand. Change one range in Table S1 and report how the reach fraction moves.

7. **Impact.** From Eq. (4), find the vertical speed that breaks a tibia for stopping distances of 10–35 cm. Estimate the vertical speed of a craft flying at V along a path 50° below horizontal, and of one that glides into the ground at its sink rate. What does each tell you about how Eilmer landed?

Exercises 2 and 6 ask students to work with quantities nobody measured: to bound them from physical arguments, sample them, and find which conclusions change when a bound moves.

7 Conclusion

A furlong's flight from the abbey tower is physically possible for a cloth wing, but only in a strong southwest wind, with a wing near the better end of what the materials allow, and on a track that follows the slope. A flight straight out over the valley does not reach that distance in any sampled case. Without a tail the craft is unstable in pitch, with a disturbance doubling in about the time a man takes to react, so a flight of some tens of seconds followed by a loss of control in a gust fits the models; a tail removes the instability. The broken legs require a fast, steep arrival, which a loss of control provides. Eilmer's own explanation of his fall is consistent with the physics, though the physics cannot show that it was the only cause.

Data and code availability

The analysis code, data manifests, and figure scripts reproduce every number and figure in this paper. They are openly available, and archived with a DOI, at

`https://github.com/silverdavi/eilmer-flight-feasibility`
`https://doi.org/10.5281/zenodo.20706321`

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